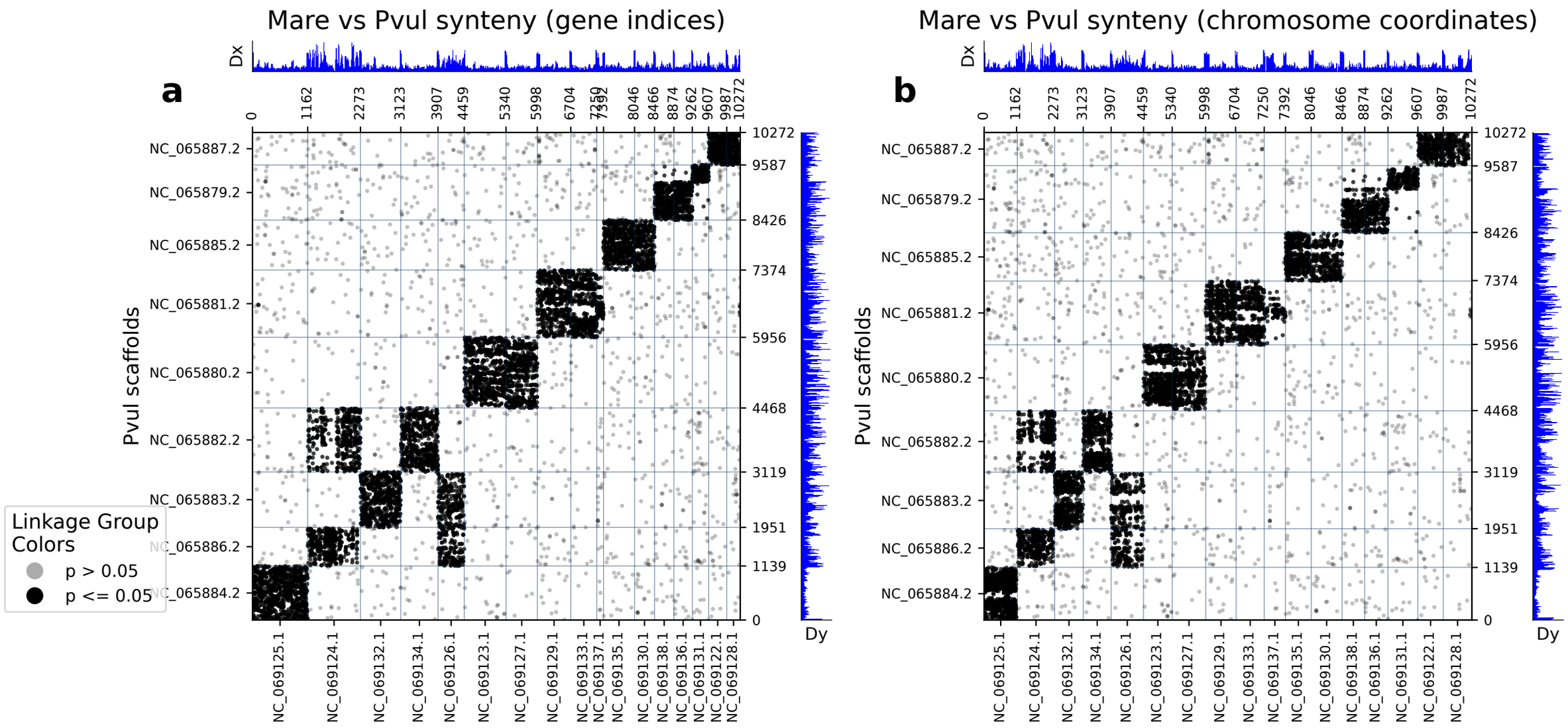
diamond: 10273 orthologs on 17 Mare scaffolds and 9 Pvul scaffolds.



c

Mare vs Pvul groupings

d

Mare vs Pvul FET, whole chroms.

	Mare scaf	Pvul scaf	p (FET)	Count
0	NC_069125.1	NC_065884.2	0.00e+00	983
1	NC_069123.1	NC_065880.2	0.00e+00	775
2	NC_069132.1	NC_065883.2	0.00e+00	732
3	NC_069134.1	NC_065882.2	0.00e+00	680
4	NC_069129.1	NC_065881.2	0.00e+00	593
5	NC_069127.1	NC_065880.2	0.00e+00	570
6	NC_069135.1	NC_065885.2	0.00e+00	554
7	NC_069124.1	NC_065886.2	3.20e-299	492
8	NC_069124.1	NC_065882.2	3.14e-163	487
9	NC_069133.1	NC_065881.2	0.00e+00	486
10	NC_069130.1	NC_065885.2	2.80e-283	345
11	NC_069138.1	NC_065879.2	4.92e-266	340
12	NC_069136.1	NC_065879.2	2.62e-251	323
13	NC_069122.1	NC_065887.2	0.00e+00	315
14	NC_069126.1	NC_065883.2	1.48e-124	281
15	NC_069131.1	NC_065879.2	4.63e-181	262
16	NC_069128.1	NC_065887.2	5.89e-197	213
17	NC_069126.1	NC_065886.2	1.14e-76	192
18	NC_069137.1	NC_065881.2	7.37e-61	107

e

Mare vs Pvul FET, chrom. pieces

	Mare scaf	Pvul scaf	p (FET)	Count
0	NC_069125.1 : 0-end	NC_065884.2 : 0-end	0.00e+00	983
1	NC_069123.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	775
2	NC_069132.1 : 0-end	NC_065883.2 : 0-end	0.00e+00	732
3	NC_069134.1 : 0-end	NC_065882.2 : 0-end	0.00e+00	680
4	NC_069129.1 : 0-end	NC_065881.2 : 0-end	0.00e+00	593
5	NC_069127.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	570
6	NC_069135.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	554
7	NC_069124.1 : 0-end	NC_065886.2 : 0-end	3.20e-299	492
8	NC_069124.1 : 0-end	NC_065882.2 : 0-end	3.14e-163	487
9	NC_069133.1 : 0-end	NC_065881.2 : 0-end	0.00e+00	486
10	NC_069130.1 : 0-end	NC_065885.2 : 0-end	2.80e-283	345
11	NC_069138.1 : 0-end	NC_065879.2 : 0-end	4.92e-266	340
12	NC_069136.1 : 0-end	NC_065879.2 : 0-end	2.62e-251	323
13	NC_069122.1 : 0-end	NC_065887.2 : 0-end	0.00e+00	315
14	NC_069126.1 : 0-end	NC_065883.2 : 0-end	1.48e-124	281
15	NC_069131.1 : 0-end	NC_065879.2 : 0-end	4.63e-181	262
16	NC_069128.1 : 0-end	NC_065887.2 : 0-end	5.89e-197	213
17	NC_069126.1 : 0-end	NC_065886.2 : 0-end	1.14e-76	192
18	NC_069137.1 : 0-end	NC_065881.2 : 0-end	7.37e-61	107

(a) depicts the chromosomes/scaffolds of Mare (x-axis) plotted against the chromosomes/scaffolds of Pvul (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10273 orthologs found between 17 Mare scaffolds and 9 Pvul scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.